MILITARY SPECIFICATION SHEET

BOLT, STRAIGHT SHANK, EXPANDABLE, 100° REDUCED FLUSH HEAD, 125 KSI Fsu (CRES 304 SLEEVE, NICKEL-ALLOY INCONEL 718, 450°F)

This specification sheet is approved for use within Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the sleevebolt described herein shall consist of this specification and the latest issue of MIL-B-85667(AS).

1. Application criteria. The sleevebolt shall be used to provide a precision net fit or controlled interference fit. The sleevebolts shall be used in accordance with the limitations of MIL-B-85667/1(AS). The sleevebolts shall not be degreased and cetyl alcohol shall not be removed before installation.

2. Companion and accessory parts. Nuts in accordance with MIL-N-85730 shall be used with the sleevebolts.

3. Oversize bolts. Second and third size sleevebolts may replace MIL-B-85667/6 sleevebolts for second and third repair operations.

4. Tooling. Wrenches shall be in accordance with MIL-W-8982.

5. Military installation. Installation shall be in accordance with MIL-HDBK-271.

6. Materials.

6.1 Pin. Pin material shall be nickel-alloy Inc 718 in accordance with AMS 5662.

6.2 Cold work. Pin shall be cold worked to 220 KSI Ftu and 125 KSI Fsu.

6.3 Sleeve. Sleeve material shall be in accordance with MIL-T-8504, AISI 304.

7. Heat treatment. Pin shall be heat treated in accordance with AMS 5662.

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7.1.1 Hardness shall not be greater than 50 HRC.

7.2 Sleeve. None (cold worked).

8. Surface texture. Surface texture shall be R_a-32 in accordance with ANSI B46.1.

9. Coatings or surface material.

9.1 Pin. The pin shall be dry-film lubricated in accordance with MIL-L-8937, .0003/.0005 thick.

9.2 Sleeve. None.

10. Lubricants. Dry film lubricant shall be applied to the pin outside diameter in accordance with MIL-L-8937 with 0.0001-inch overspray on head and threads. After application of dry lubrication the pin inside diameter shall be lubricated with cetyl alcohol in accordance with MIL-L-87132, or BOELUBE.

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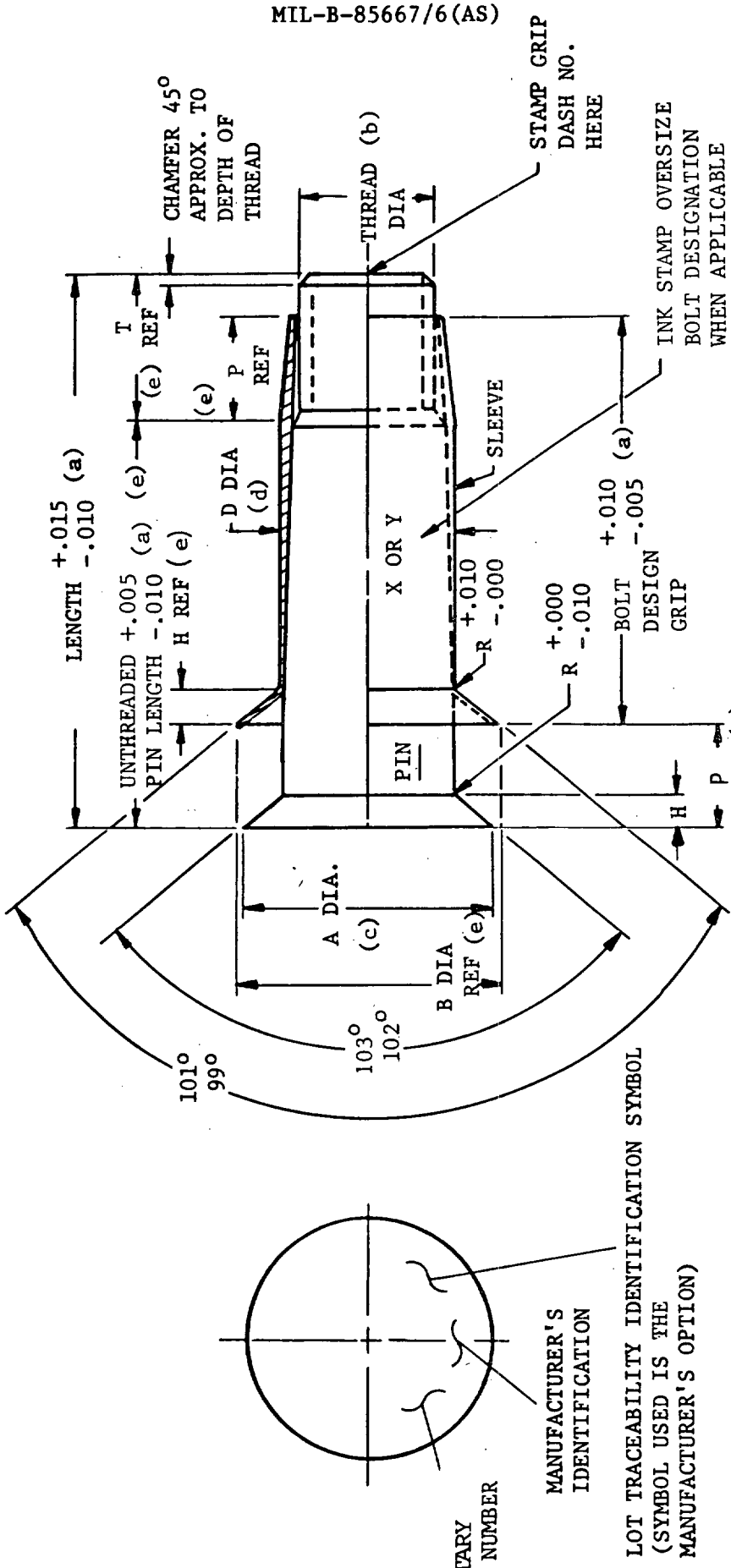


FIGURE 1. Reduced flush head bolt.

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NOTES:

a. Grip lengths are available in .0625 increments.

b. Threads shall be in accordance with MIL-S-8879. Thread major diameter (T.D.) shall be in accordance with Table I.

c. 1. For sizes .190 thru .375, dimension "A" shall be concentric to dimension "D" within .008 T.I.R.

2. For sizes .437 thru .625, .012 T.I.R.

3. For sizes .750 thru 1.000, .020 T.I.R.

d. 1. For sizes .190 thru .375, dimension "D" shall be concentric to thread pitch diameter within .006 T.I.R.

2. For sizes .437 thru .625, .010 T.I.R.

3. For sizes .750 thru 1.000, .016 T.I.R.

e. Reference dimensions are for design purposes only and are not an inspection or manufacturing requirement.

f. For grip numbers -6 thru -8 (group 1), taper ratios (reference only) are 25:1 and 48:1 for groups II and III (see Table I).

g. Protrusion will vary according to taper to provide expansion as given in Table I.

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TABLE I. Bolt dimensions.

SIZE NO'S	GROUP	GRIP DASH NUMBERS	THREAD PER MIL-S-8879 UNFJ-3A	A BOLT HEAD DIA.	B REF SLEEVE FLANGE DIAMETER			H	FH FLANGE HEIGHT REF.	P (g)	R FILLET RADIUS	T.D. THREAD DIA.	T THREAD LENGTH	D DIMENSION INITIAL SLEEVE DIAMETER (a)			D' DIMENSION FINAL SLEEVE DIAMETER (a) (b)			MINIMUM INSTALLATION HOLE SIZE DIAMETER		
					STD.	X O/S	Y O/S							STD.	X O/S	Y O/S	STD.	X O/S	Y O/S	STD.	X O/S	Y O/S
-3	1	-6 THRU -8	.1900-32	.3327 .3259	.3820	.3880	.3940	.048 .046	.061	.225 .215	.030	.1840 .1810	.343	.2365 .2360	.2425 .2420	.2485 .2480	.2455 .2445	.2515 .2505	.2575 .2565	.237	.243	.249
	2	-9 THRU -16		.3341 .3273	.3838	.3898	.3958		.062	.432 .422				.2365 .2360	.2425 .2420	.2485 .2480	.2455 .2445	.2515 .2505	.2575 .2565	.237	.243	.249
	3	-17 THRU -24		.3445 .3377	.4004	.4074	.4144		.064	.432 .422				.2485 .2480	.2555 .2550	.2625 .2620	.2575 .2565	.2645 .2635	.2715 .2705	.249	.256	.263
-4	1	-6 THRU -8	.2500-28	.4295 .4224	.4772	.4882	.4942	.063 .061	.074	.225 .215	.030	.2440 .2410	.406	.3005 .3000	.3115 .3110	.3175 .3170	.3095 .3085	.3205 .3195	.3265 .3255	.301	.312	.318
	2	-9 THRU -17		.4339 .4268	.4805	.4915	.4975		.075	.432 .422				.3005 .3000	.3115 .3110	.3175 .3170	.3095 .3085	.3205 .3195	.3265 .3255	.301	.312	.318
	3	-18 THRU -28		.4469 .4398	.5011	.5081	.5171		.078	.432 .422				.3145 .3140	.3215 .3210	.3305 .3300	.3235 .3225	.3305 .3295	.3395 .3385	.315	.322	.331
-5	1	-6 THRU -8	.3125-24	.5080 .5013	.5556	.5626	.5696	.070 .068	.080	.225 .215	.040	.3060 .3020	.484	.3665 .3660	.3735 .3730	.3805 .3800	.3755 .3745	.3825 .3815	.3895 .3885	.367	.374	.381
	2	-9 THRU -20		.5163 .5090	.5630	.5700	.5770		.082	.432 .422				.3665 .3660	.3735 .3730	.3805 .3800	.3755 .3745	.3825 .3815	.3895 .3885	.367	.374	.381
	3	-21 THRU -31		.5306 .5233	.5836	.5916	.5986		.086	.432 .422				.3805 .3800	.3885 .3880	.3955 .3950	.3895 .3885	.3975 .3965	.4045 .4035	.381	.389	.396
-6	1	-6 THRU -8	.3750-24	.5977 .5876	.6434	.6514	.6634	.081 .078	.090	.225 .215	.040	.3680 .3640	.562	.4315 .4310	.4395 .4390	.4515 .4510	.4405 .4395	.4485 .4475	.4605 .4595	.432	.440	.452
	2	-9 THRU -22		.6087 .5986	.6536	.6616	.6736		.093	.432 .422				.4315 .4310	.4395 .4390	.4515 .4510	.4405 .4395	.4485 .4475	.4605 .4595	.432	.440	.452
	3	-23 THRU -34		.6243 .6142	.6754	.6834	.6914		.096	.432 .422				.4475 .4470	.4555 .4550	.4635 .4630	.4565 .4555	.4645 .4635	.4725 .4715	.448	.456	.464
-7	1	-6 THRU -8	.4375-20	.7072 .6968	.7571	.7691	.7771	.100 .097	.108	.225 .215	.050	.4310 .4260	.625	.4985 .4980	.5105 .5100	.5185 .5180	.5075 .5065	.5195 .5185	.5275 .5265	.499	.511	.519
	2	-9 THRU -26		.7234 .7130	.7729	.7849	.7929		.115	.432 .422				.4985 .4980	.5105 .5100	.5185 .5180	.5075 .5065	.5195 .5185	.5275 .5265	.499	.511	.519
	3	-27 THRU -40		.7416 .7312	.7974	.8084	.8204		.117	.432 .422				.5185 .5180	.5295 .5290	.5415 .5410	.5275 .5265	.5385 .5375	.5505 .5495	.519	.530	.542
-8	1	-6 THRU -8	.5000-20	.7937 .7831	.8425	.8525	.8625	.110 .107	.114	.225 .215	.050	.4930 .4880	.703	.5695 .5690	.5795 .5790	.5895 .5890	.5785 .5775	.5885 .5875	.5985 .5975	.570	.580	.590
	2	-9 THRU -30		.8157 .8051	.8637	.8737	.8837		.123	.432 .422				.5695 .5690	.5795 .5790	.5895 .5890	.5785 .5775	.5885 .5875	.5985 .5975	.570	.580	.590
	3	-31 THRU -46		.8366 .8260	.8908	.9008	.9098		.127	.432 .422				.5895 .5890	.5995 .5990	.6085 .6080	.5985 .5975	.6085 .6075	.6175 .6165	.590	.600	.609

NOTES:

(a) "D" DIA. SHALL BE VERIFIED WITHIN ITS GRIP BEARING PORTION BEFORE EXPANSION; WHEN SLEEVE IS PUSHED TO BOLT HEAD, THE "D" DIA. SHALL BE WITHIN THE EXPANDED "D" DIA. LIMITS.

(b) THIS BOLT IS MANUFACTURED AS A PRECISION ASSEMBLY. REMOVAL OF SLEEVE FROM PIN BY USER WILL RESULT IN LOSS OF THE BOLT'S ABILITY TO MEET MIN/MAX EXPANSION LIMITS.

(c) ALL DIMENSIONS ARE IN INCHES. TOLERANCES: DECIMALS ±.010, ANGLES ±2°.

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TABLE II. Bolt weights.

GRIP NUMBER	BOLT DESIGN GRIP	-3 DIAMETER				-4 DIAMETER				-5 DIAMETER			
		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100	
				PIN	BOLT			PIN	BOLT			PIN	BOLT
6	.375	.405	.748	.3411	.3810	.405	.811	.6297	.6889	.405	.889	1.0442	1.1265
7	.438	.468	.811	.3706	.4185	.468	.874	.6798	.7504	.468	.952	1.1212	1.2488
8	.500	.530	.873	.3989	.4553	.530	.936	.7282	.8108	.530	1.014	1.1959	1.3096
9	.562	.592	.935	.4449	.4986	.592	.998	.8047	.8824	.592	1.076	1.3244	1.4208
10	.625	.655	.998	.4760	.5373	.655	1.061	.8573	.9449	.655	1.139	1.4065	1.5151
11	.688	.718	1.061	.5066	.5735	.718	1.124	.9093	1.0071	.718	1.202	1.4879	1.6091
12	.750	.780	1.123	.5634	.6127	.780	1.186	.9600	1.0684	.780	1.264	1.5673	1.7016
13	.812	.842	1.185	.5658	.6499	.842	1.248	1.0102	1.1294	.842	1.326	1.6461	1.7938
14	.875	.905	1.248	.5680	.6603	.905	1.311	1.0608	1.1913	.905	1.389	1.7256	1.8872
15	.938	.968	1.311	.6244	.7250	.968	1.374	1.1108	1.2528	.968	1.452	1.8046	1.9662
16	1.000	1.030	1.373	.6526	.7618	1.030	1.436	1.1595	1.3134	1.030	1.514	1.8815	2.0721
17	1.062	1.092	1.435	.7384	.8687	1.092	1.498	1.2077	1.3738	1.092	1.576	1.9578	2.1635
18	1.125	1.155	1.498	.7693	.9098	1.155	1.561	1.3658	1.5513	1.155	1.639	2.0348	2.2561
19	1.188	1.218	1.561	.7999	.9505	1.218	1.624	1.4195	1.6184	1.218	1.702	2.1112	2.3484
20	1.250	1.280	1.623	.8293	.9903	1.280	1.686	1.4718	1.6844	1.280	1.764	2.1856	2.4392
21	1.312	1.234	1.685	.8591	1.0309	1.234	1.748	1.5235	1.7502	1.342	1.826	2.4168	2.6950
22	1.375	1.405	1.748	.8886	1.0714	1.405	1.811	1.5756	1.8167	1.405	1.889	2.4983	2.7943
23	1.438	1.468	1.811	.9177	1.1117	1.468	1.874	1.6273	1.8831	1.468	1.952	2.5791	2.8932
24	1.500	1.530	1.873	.9456	1.1511	1.530	1.936	1.6775	1.9484	1.530	2.014	2.6580	2.9907
25	1.562					1.592	1.998	1.7272	2.0135	1.592	2.076	2.7362	3.0875
26	1.625					1.655	2.061	1.7773	2.0794	1.655	2.139	2.8152	3.1862
27	1.688					1.718	2.124	1.8269	2.1451	1.718	2.702	2.8935	3.2843
28	1.750					1.780	2.186	1.8751	2.2097	1.780	2.264	2.9698	3.3809
29	1.812									1.842	2.326	3.0455	3.4772
30	1.875									1.905	2.389	3.1220	3.5747
31	1.938									1.968	2.452	3.1978	3.6720

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TABLE II. Bolt weights (continued).

GRIP NUMBER	BOLT DESIGN GRIP	-6 DIAMETER				-7 DIAMETER				-8 DIAMETER			
		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100		UNTHREADED PIN LENGTH	BOLT LENGTH	WEIGHT/100	
				PIN	BOLT			PIN	BOLT			PIN	BOLT
6	.375	.405	.967	1.6283	1.7349	.405	1.030	2.3649	2.5038	.405	1.108	3.3195	3.5127
7	.438	.468	1.030	1.7381	1.8641	.468	1.093	2.5139	2.6772	.468	1.171	3.5128	3.7386
8	.500	.530	1.092	1.8797	2.0260	.530	1.155	2.6588	2.8475	.530	1.233	3.7012	3.9609
9	.562	.592	1.154	2.0312	2.1478	.592	1.217	2.9335	3.0653	.592	1.295	4.0790	4.2522
10	.625	.655	1.217	2.1485	2.2797	.655	1.280	3.0948	3.2430	.655	1.358	4.2906	4.4845
11	.688	.718	1.280	2.2651	2.3963	.718	1.343	3.2552	3.4240	.718	1.421	4.5011	4.7158
12	.750	.780	1.342	2.3790	2.5254	.780	1.405	3.4121	3.5950	.780	1.483	4.7071	4.9440
13	.812	.842	1.404	2.4921	2.6702	.842	1.467	3.5681	3.7692	.842	1.545	4.9122	5.1720
14	.875	.905	1.467	2.6063	2.8011	.905	1.530	3.7260	3.9459	.905	1.608	5.1195	5.4029
15	.938	.968	1.530	2.7199	2.9318	.968	1.593	3.8827	4.1220	.968	1.671	5.3259	5.6335
16	1.000	1.030	1.592	2.8307	3.0426	1.030	1.655	4.0355	4.2947	1.030	1.733	5.3279	5.8604
17	1.062	1.092	1.654	2.9404	3.1700	1.092	1.717	4.1881	4.4679	1.092	1.795	5.7289	6.0870
18	1.125	1.155	1.717	3.0521	3.3184	1.155	1.780	4.3422	4.6431	1.155	1.858	5.9322	6.3166
19	1.188	1.218	1.780	3.1621	3.4525	1.218	1.843	4.4955	4.8181	1.218	1.921	6.1345	6.5458
20	1.250	1.280	1.842	3.2699	3.5749	1.280	1.905	4.6453	4.9902	1.280	1.983	6.3325	6.7713
21	1.312	1.342	1.904	3.3771	3.7022	1.342	1.967	4.7943	5.1621	1.342	2.045	6.5287	6.9958
22	1.375	1.405	1.967	3.4853	3.8310	1.405	2.030	4.9449	5.3361	1.405	2.108	6.7280	7.2239
23	1.438	1.468	2.030	3.8147	4.1995	1.468	2.093	5.0947	5.5099	1.468	2.171	6.9262	7.4517
24	1.500	1.530	2.092	3.9285	4.3358	1.530	2.155	5.2411	5.6563	1.530	2.233	7.1202	7.6759
25	1.562	1.592	2.154	4.0415	4.4719	1.592	2.217	5.3866	5.8264	1.592	2.295	7.3131	7.8997
26	1.625	1.655	2.217	4.1558	4.6097	1.655	2.280	5.5338	5.9988	1.655	2.358	7.5084	8.1265
27	1.688	1.718	2.280	4.2693	4.7473	1.718	2.343	6.0341	6.5248	1.718	2.421	7.7027	8.3529
28	1.750	1.780	2.342	4.2510	4.7535	1.780	2.405	6.1891	6.7913	1.780	2.483	7.8927	8.5757
29	1.812	1.842	2.404	4.4901	5.0176	1.842	2.467	6.3433	6.9343	1.842	2.545	8.0818	8.7983
30	1.875	1.905	2.467	4.6014	5.1545	1.905	2.530	6.4993	7.1197	1.905	2.608	8.2731	9.0237
31	1.938	1.968	2.530	4.7119	5.2910	1.968	2.593	6.6543	7.3047	1.968	2.671	8.9917	9.7919
32	2.000	2.030	2.592	4.8198	5.4254	2.030	2.655	6.8059	7.4568	2.030	2.733	9.1936	10.0301
33	2.062	2.092	2.654	4.9269	5.5595	2.092	2.717	6.9566	7.6686	2.092	2.795	9.3945	10.2679
34	2.125	2.155	2.717	5.0351	5.6951	2.155	2.780	7.1090	7.8527	2.155	2.858	9.5979	10.5090
35	2.188					2.218	2.843	7.2605	8.0374	2.218	2.921	9.8002	10.7495
36	2.250					2.280	2.905	7.4086	8.2174	2.280	2.983	9.9981	10.9864
37	2.312					2.342	2.967	7.5558	8.3981	2.342	3.045	10.1949	11.2220
38	2.375					2.405	3.030	7.7047	8.5783	2.405	3.108	10.3942	11.4623
39	2.438					2.468	3.093	7.8528	8.7988	2.468	3.171	10.5926	11.7017
40	2.500					2.530	3.155	7.9975	8.9792	2.530	3.233	10.7865	11.9371
41	2.562									2.592	3.295	10.9794	12.1723
42	2.625									2.655	3.358	11.1747	12.4104
43	2.688									2.718	3.421	11.3973	12.6266
44	2.750									2.780	3.483	11.5590	12.8825
45	2.812									2.842	3.545	11.7479	13.1162
46	2.875									2.905	3.608	11.9394	13.3532

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TABLE III. Mechanical properties.

Diameter Dash Number	Double Shear Strength Lbs-Min (a)	Double Shear Nominal Test	Pin Test Dia Area (a)	Max Install Torque Inch Lbs
-3	7,090	.1900	.0284	40
-4	12,300	.2500	.0491	85
-5	19,200	.3125	.0767	140
-6	27,800	.3750	.1104	200
-7	37,500	.4375	.1503	300
-8	49,200	.5000	.1963	455

Notes:

a. Shear test performed on pins ground to pin test diameter. The shear values are based on a minimum single shear strength of 125 KSI Fsu and the stress area at nominal test diameter.

b. Tensile fatigue and stress durability tests not applicable to 100° reduced flush head alloy.

11. Example of military part number.

M85667/6 - 3 - 34

- 34 — Grip length dash number in sixteenths of an inch. (Stamp on threaded end.)
- '-' — designates basic diameter: X designates 1st oversize, Y designates 2nd oversize. (Stamp on sleeve.)
- 3 — Diameter dash numbers in sixteenths of an inch.
- M85667/6 — Basic specification sheet number.

12. Administrative and contractual provisions.

a. For design purposes and in case of conflict, this document takes precedence over documents referenced herein.

b. Definitions of terms used in this document are in accordance with MIL-STD-1515.

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c. For additional specification sheets and standards, see MIL-B-85667(AS), Supplement 1.

d. The bolt assemblies identified herein are proprietary products and are covered by the following patents: U.S. 4,087,896 (expiring May 9, 1997); 4,106,036 (expiring July 25, 1997); 4,048,898 (expiring September 20, 1996); Great Britain 1,524,886; Canada 1,023,585; Japan 50-125256; Italy 1,051,593; Israel 515578; and Germany 2,546,641.5. The Government does not have a royalty-free license.

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